



Live radio from anywhere

You can listen to radio stations from around the world through an interactive world map from your browser. <https://digit.in/jan19-93>



China cautions researchers

Chinese researchers warned to remove all sensitive information from devices before travelling to the US. <https://digit.in/jan19-94>

themselves from the very early days, the origin of the term may go back to before the invention of radio itself. Among telegraph operators, the words “ham” and “plug” were both used to describe “a telegraph operator, who lacks ability,” according to The telegraph instructor by George M Dodge. There is speculation that the word may have originated from the phrase “ham fisted”, which indicates the clumsiness when using a piece of equipment.

The stations by the amateurs often outstripped the capabilities of the stations built by the Navy, and in fact, the amateurs were happy to help the Navy send and receive messages in emergency situations. One of the early uses of radio communications was for dispatching help to ships in the sea that needed it, and co-ordinating the rescue operations with the vessels that followed up. The amateur radio stations around Boston interfered with the rescue operations of the RMS Republic, which collided with the SS Florida in 1909. However, because of a Marconi Wireless Transmitter on board, about 1,500 passengers on the Republic were saved. Many more lives could potentially have been saved if radio technologies would have been used to their fullest potential when the Titanic sank in 1912. As a consequence of an investigation into the sinking, the Radio Act of 1912 was passed in the US, restricting amateur radio operations to “shortwave” frequencies of 1500 kHz or higher. This was one of the earliest laws to regulate radio communication, and subsequently similar laws would be introduced around the world.

PIRATE RADIO

One of the problems with radio transmission was the static and interference because of the electrical equipment itself, and external sources such as thunderstorms. Amateur radio operators in the 30s were using a frequency modulation (FM) approach to reduce the noise. In 1933, Edwin H Armstrong, an electrical engineer, developed a wideband FM

system. A series of tests demonstrated the capabilities of reducing noise, and transmitting high fidelity signals. Between 1930s and the 1940s, television and audio programming began to be broadcasted over radio.

The ether became increasingly regulated, and in the UK, the BBC was given exclusive rights for commercial broadcasts, and to charge for advertisements – a monopoly that lasted till 1973.

Radio Luxembourg, situated in mainland Europe began to broadcast commercial English content aimed at England and Ireland in 1933. Although the British authorities complained because of the unauthorised use of airspace, there is nothing that they could really do about it. The station quickly became popular because of the popular music that they played, as well as broadcasting at hours when the BBC was inactive. Many British companies openly advertised on the station. Radio Luxembourg was a precursor to “pirate” radio stations, that began to broadcast commercial content without having the license to do so.

Inspired from Radio Luxembourg, Radio Mercur, a vessel began broadcasting radio content to Denmark and Sweden in 1958. From the 60s to the 80s, many pirate radio stations sprung up delivering content from almost every country in Europe, broadcasting from offshore vessels.

RADIO SATELLITES

The first satellite for ham communications, OSCAR was deployed only four years after the launch of Sputnik in 1957, the first ever satellite. OSCAR was the first passenger satellite to piggyback on another mission, with the primary payload being the Discoverer 26. The orbit of OSCAR decayed over 22 days, but was a success among radio operators within that short duration. The earliest amateur radio satellites were built with off the shelf components available from electronic stores. On board are repeaters that receive radio signals and then retransmit it, and transponders for shifting com-

munications between the uplink and downlink frequencies. Amateur radio satellites also have a store and forward capability, which acts like a bulletin board for ham operators around the world. Messages received in one location can be transmitted in another location. There are over a hundred active satellites for licensed amateur radio operators in orbit. There have been over 200 satellites launched with payloads for the benefit of ham operators so far, built by 23 countries.

India is one of the many countries that have launched satellites that provide services to the national and international amateur radio community. HAMSAT was one of the OSCAR series of satellites, and was launched by ISRO in 2005. It was popular because of the sensitivity of its receiver, and the power of its transmitter. Although designed for a mission duration of one year, the satellite was in service for 10 years. Two academic satellites launched by ISRO and developed by Indian institutions had store and forward capabilities for the benefit of the international ham radio community, the Anna University Satellite (ANUSAT) launched in 2009 and the Swayam launched in 2016. The College of Engineering, Pune is developing the Swayam II. Exseed Sat-1, built by Mumbai based Exseedspace launched a satellite on board the SpaceX Falcon 9 on 5 December 2018, to improve the range of ham radio operators. The satellite has been hacked together, and includes an Arduino, some fishing line and a carpenter's measuring tape that serves as an antenna.

Today, those who are interested in becoming ham operators can get a license in India by passing a test. Local radio clubs can help, but it is possible to study using material available freely on the internet. The syllabus covers principles of radio, the regulations for radio operations in India, and the ability to send and receive Morse code messages at speeds in excess of 8 words per minute. The National Institute of Amateur Radio website (niar.org) is a good place to start. 